AI-Powered Multi-Disease Prediction System

Abstract

The AI-Powered Multi-Disease Prediction System is a machine learning-based web application designed to predict multiple diseases, including Diabetes, Heart Disease, Liver Disease, and Kidney Disease. The system uses patient health data and trained machine learning models to provide quick and accurate predictions. It helps users identify potential health risks and receive basic health recommendations.

Objective

- Predict multiple diseases using machine learning.

- Provide a simple and user-friendly interface.

- Generate instant health predictions.

- Assist in early disease detection.

Technologies Used

- Python

- Pandas

- NumPy

- Scikit-Learn

- Streamlit

- Joblib

- GitHub

Methodology

1. Collect healthcare datasets.

2. Clean and preprocess the data.

3. Train machine learning models using Random Forest Classifier.

4. Evaluate model performance.

5. Develop a web application using Streamlit.

6. Generate disease predictions based on user input.

Features

- Diabetes Prediction
- Heart Disease Prediction
- Liver Disease Prediction
- Kidney Disease Prediction
- Health Recommendations
- Interactive Web Interface

Result

The system successfully predicts disease risks using medical parameters and provides users with quick and reliable results. The trained models achieve good accuracy and help in early health assessment.

Conclusion

This project demonstrates the application of machine learning in healthcare. It provides an efficient and user-friendly solution for disease prediction and can be further enhanced with additional diseases and real-time healthcare integration.